ASL research retrospective

Past efforts and key results

Research through October 19, 2026 · Prepared October 20, 2026

Our ASL work has progressed from building an end-to-end prototype to identifying why apparently fluent translations fail, establishing more reliable evaluation, and testing specialized models. The clearest progress is in understanding the bottlenecks: **the visual representation and the alignment between signing and text matter substantially; simply adding frames, training data, or predicted concepts has not reliably improved translation.**

We have not yet demonstrated reliable translation of customer uploads or a fusion model that improves on Uni-Sign. This account follows the ASL index's October 16 revision, with the October 19 follow-up identified separately. Sources are linked at the end.

1. Initial prototype: the product pipeline worked before recognition quality did

The initial goal was ASL video to English text, reusing Curify's existing translation, subtitle, voice-over, and video-generation infrastructure.

A general-purpose Gemini model enabled an end-to-end demo, but evaluations exposed a serious distinction between producing plausible English and understanding signing. Outputs could be fluent, unrelated to the video, and inconsistent across runs.

Engineering improvements addressed frame budgets, repeatability, duplicate segments, configuration handling, and unverified-output notices. These made the system easier to operate and inspect, but did not establish recognition accuracy.

Key result: completing the pipeline was an engineering achievement; accurate ASL recognition remained the unsolved component.

2. Early evaluation exposed contamination and weak model compatibility

In August, the team tried the TGCN-WLASL isolated-sign recognizer with a MediaPipe-based input adapter. Both tested model variants scored **0/12 top-1 and top-5** on a small set of known WLASL examples. Hand swapping, mirroring, normalization, and scaling changes did not recover performance.

The findings pointed toward an input-representation or checkpoint compatibility problem. They did not conclusively isolate the cause, because a successful comparison using the original OpenPose inputs was still needed. The exercise also clarified that isolated-sign classification is a different task from continuous sentence translation.

Separately, the team discovered contamination in early evaluation material and retracted affected claims. Subsequent work increasingly emphasized independent references, frozen inputs, and traceable predictions.

Key result: model loading and plausible demos were insufficient evidence. Input compatibility and evaluation provenance became essential checks.

3. September benchmark work showed that simple sampling changes did not recover meaning

On a 128-clip How2Sign benchmark, the Gemini-based pipeline produced only about **41% of the reference word count**. Deletions accounted for roughly **61% of edit operations**, alongside substitutions and unrelated content.

The team tested several explanations:

Intervention	Result	Interpretation
Increase sampling from 2 to 8 fps	On 76 paired clips, WER changed from 0.9730 to 0.9715	Essentially flat on the completed subset
Replace motion-weighted sampling with uniform sampling	No useful improvement on 12 diagnostic clips	Sampling strategy alone did not resolve the failure
Use overlapping frame batches	WER improved, but chrF declined and all 12 remained topic-mismatched	Shorter output could improve one metric without improving meaning
Send native video instead of sampled frames	An encouraging single example did not reproduce: 0/5 successfully evaluated diagnostic clips recovered meaning; seven failed before inference	The broader test remained incomplete
Supply tracked trajectories as text	No meaning recovery in five tested clips	Extracting motion did not make the VLM understand it

Key result: the problem could not be explained solely by too few frames or insufficient overlap. The work also showed why WER, text similarity, and output length must be interpreted together.

4. Uni-Sign became a stronger research baseline, but did not establish customer readiness

The team then ran the released Uni-Sign checkpoint end to end. It provided a stronger benchmark baseline and a purpose-built sign representation.

However, the tested real-user video still produced severe repetition and degeneration. The implementation also exposed a long-input issue: inputs above its 256-frame limit could undergo random subsampling. Fixed or overlapping windows addressed aspects of that mechanism without demonstrating a corresponding translation-quality improvement.

Service integration and an opt-in backend adapter were developed, but these were distinct from validating the model on customer uploads.

Key result: Uni-Sign became the strongest tested research foundation in this workstream. Its benchmark advantage did not establish that it was ready to replace the customer-facing recognizer.

5. October training experiments separated data scale from representation quality

Two controlled experiments clarified why the custom translator struggled.

P0-11: more supervised data did not reliably improve the landmark model. With matched initialization and a fixed training-step budget, expanding from 2,751 to 26,419 clips produced no statistically significant BLEU improvement. At larger sizes, predictions became less diverse, and shuffled-video tests indicated weak dependence on the video input.

This result applies to the tested architecture and training recipe. Because compute was fixed, larger datasets received fewer passes; it does not establish that more pretraining data is unhelpful.

P0-13: a pretrained sign representation helped substantially.

Input to the custom downstream translator	BLEU	chrF	Distinct predictions
MediaPipe landmarks	1.02	11.12	102/147
Frozen Uni-Sign features	3.25	17.05	147/147

The BLEU improvement was **+2.23**, with a reported 95% confidence interval of **[1.28, 3.39]**. Dependence on the video also increased.

Key result: pretrained sign features improved our weak custom baseline. This was not evidence that we trained a model better than Uni-Sign itself; pretraining, architecture, and other differences were not fully isolated.

6. Preprocessing experiments identified a large, recoverable input gap

P0-12 held the Uni-Sign checkpoint and inference environment constant while changing its inputs:

Input preparation, same 128 clips	BLEU-4
Our extractor, original timestamps, full frame	5.47
Add aspect/coordinate correction	5.80
Our extractor, official timing and signer crop	8.61
Official poses and preprocessing	9.13

Matching the official preparation recovered **85.7% of the measured gap** between our original path and the official-input result.

Key result: much of the apparent model deficit came from input preparation. That experiment did not separate timing from cropping, nor show that official alignment could be reproduced automatically on customer videos.

The **October 19 follow-up**, not yet incorporated into the index, narrowed this diagnosis. On 2,347 evaluable clips:

Preparation	BLEU-4	WER, lower is better
Original timing, full frame	4.85	101.1
Original timing, automatic crop	5.24	100.9
Automatically widened timing and crop	5.77	109.7
Official realigned timing and automatic crop	13.15	87.2

Automatic cropping offered a small benefit. Widening the window increased text overlap but worsened WER. Correct timing produced a much larger gain. **P0-18 stopped at development with a NOT SUPPORTED verdict**, preserving H18 for a stronger candidate. These remain development findings, not confirmatory results. See Source 2.

Metric note: the September WER values above use the 0–1 scale; this table reports WER in percentage units. Results from different datasets, subsets, or evaluation conditions should not be compared directly.

7. Fusion found complementary information, but could not use it reliably

P0-14 tested whether Uni-Sign's internal features contained concepts omitted from its translation. They did: probes on mT5 encoder states recovered some missed concepts. However, at the tested operating point, they recovered **90** positive mentions while adding **581 false positives**.

P0-15 then trained a verifier to filter additions. Its ranking improved, but it did not meet the predefined precision requirement. A naive union added **50 correct concepts and 281 incorrect ones**; the guarded approach abstained.

There were also evaluation-integrity qualifications: a sanity-gate inconsistency limited P0-14's interpretation, and a previously observed split was retired from confirmatory use.

Key result: complementary information exists, but safe concept addition was not demonstrated. No fused sentence-translation improvement was measured. This rejects the tested approach at its operating point, rather than all possible fusion architectures.

8. Error analysis is more structured, but linguistic verification remains unfinished

The team produced automated taxonomies, concept-level evaluations, inspectors, and a 45-clip manual-review package covering names, fingerspelling, numbers, negation, questions, spatial relations, and general semantics.

These are useful diagnostic tools, but the linguistic labels remain unverified. English token checks cannot establish what was actually signed, and a selected diagnostic sample cannot estimate population error rates.

The repetition audit adds a clearer finding: on the 128-clip official-input benchmark, repetition occurred in **3/45 short, 6/50 medium, and 9/33 long clips**. Length is associated with higher risk, but short inputs also fail. The audit does not by itself isolate the decoder as the cause. See Source 3.

What this work has established

The research has produced reusable evaluation infrastructure, reproducible model execution, a stronger representation baseline, and a much clearer account of failure. It has also eliminated several attractive but unsupported explanations.

The strongest evidence now supports three priorities:

- **Solve temporal alignment:** determine which signing interval corresponds to each target utterance, rather than uniformly enlarging windows.
- **Separate input failures from generation failures:** evaluate repetition under fixed inputs and controlled decoding, while measuring omissions and semantic preservation.
- **Complete ASL-fluent video review:** use verified examples to decide whether the next investment belongs in alignment, hand/face representations, or decoding.

The central unresolved question is whether these improvements can produce **reliable meaning on unseen customer videos**, beyond better scores on a repeatedly examined benchmark.

Sources and scope

1. ASL index: October 16 committed revision. Historical findings and qualifications, particularly sections 11–18 and the appendices.
2. October 19 consolidated work report. Final P0-18 development results and review status.
3. Repetition audit. Text-rule repetition labels and official-input duration groups.

This retrospective summarizes recorded experiments; it does not report a new model run or completed ASL-fluent video review. The October 19 findings supplement the index rather than silently replacing its dated record.